Capacitance and Dielectrics

Introduction

- In flash photography, the energy used to make the flash is stored in a **capacitor**, which consists of two closely spaced conductors that carry opposite charges.
- The energy of a capacitor is actually stored in the electric field.

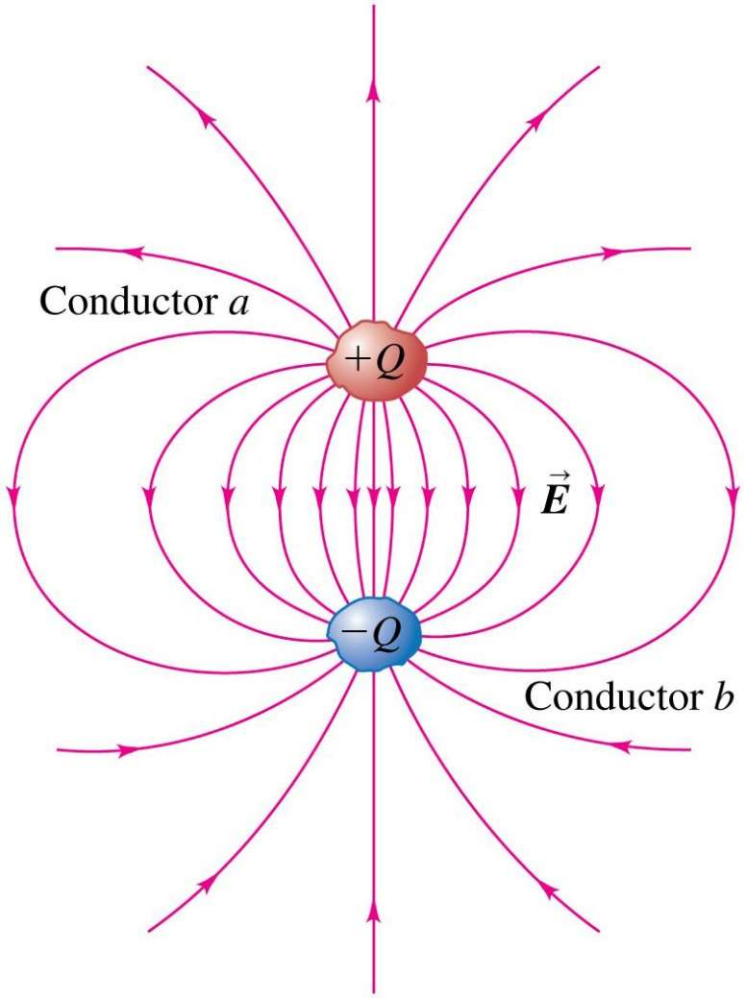


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Capacitors

- Any two conductors separated by an insulator (or a vacuum) form a **capacitor**.
- When the capacitor is **charged**, it means the two conductors have charges with equal magnitude and opposite sign, and the net charge on the capacitor as a whole is zero.



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Capacitors and Capacitance

- One common way to charge a capacitor is to connect the two conductors to opposite terminals of a battery.
- This gives a potential difference V_{ab} between the conductors that is equal to the voltage of the battery.
- If we change the magnitude of charge on each conductor, the potential difference between conductors changes; however, the **ratio** of charge to potential difference does not change.
- This ratio is called the **capacitance** C of the capacitor:

Capacitance of a capacitor

$$C = \frac{Q}{V_{ab}}$$

Q: Magnitude of charge on each conductor

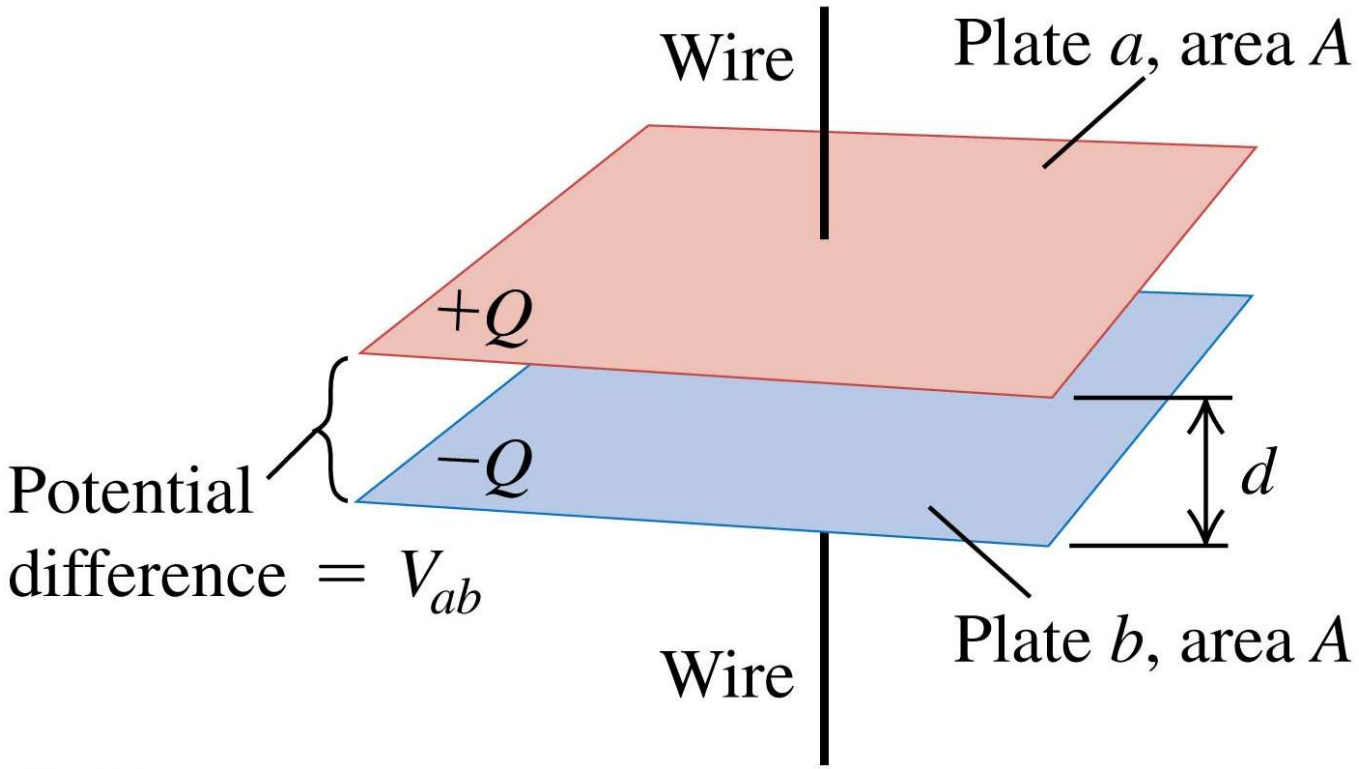
V_{ab}: Potential difference between conductors (a has charge $+Q$, b has charge $-Q$)

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Parallel-Plate Capacitor (1 of 2)

- A **parallel-plate** capacitor consists of two parallel conducting plates separated by a distance that is small compared to their dimensions.

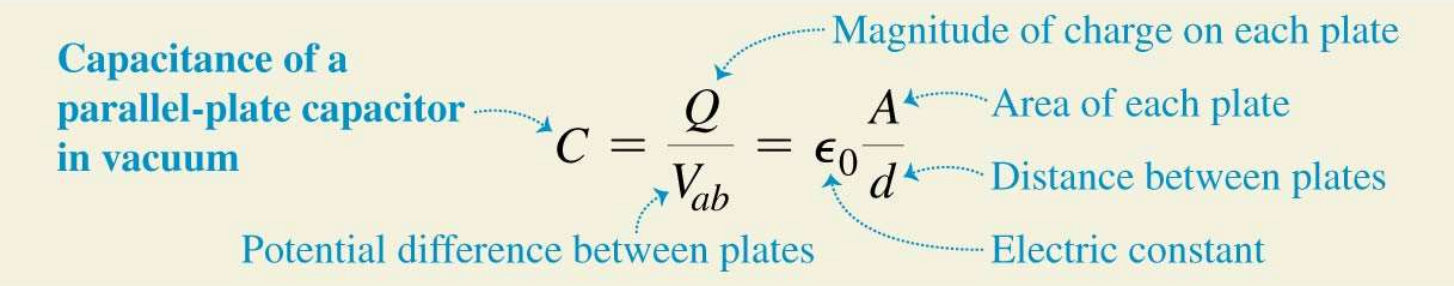


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Parallel-Plate Capacitor (2 of 2)

- The field between the plates of a parallel-plate capacitor is essentially **uniform**, and the charges on the plates are uniformly distributed over their opposing surfaces.
- When the region between the plates is empty, the capacitance is:



- The capacitance depends on only the geometry of the capacitor.
- The quantities A and d are constants for a given capacitor, and ϵ_0 is a universal constant.
- Video Tutor Solution: Example 24.2

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Condenser Microphones

- Inside a **condenser microphone** is a capacitor with one rigid plate and one flexible plate.
- The two plates are kept at a constant potential difference.
- Sound waves cause the flexible plate to move back and forth, varying the capacitance *C* and causing charge to flow to and from the capacitor.
- Thus a sound wave is converted to a charge flow that can be amplified and recorded digitally.



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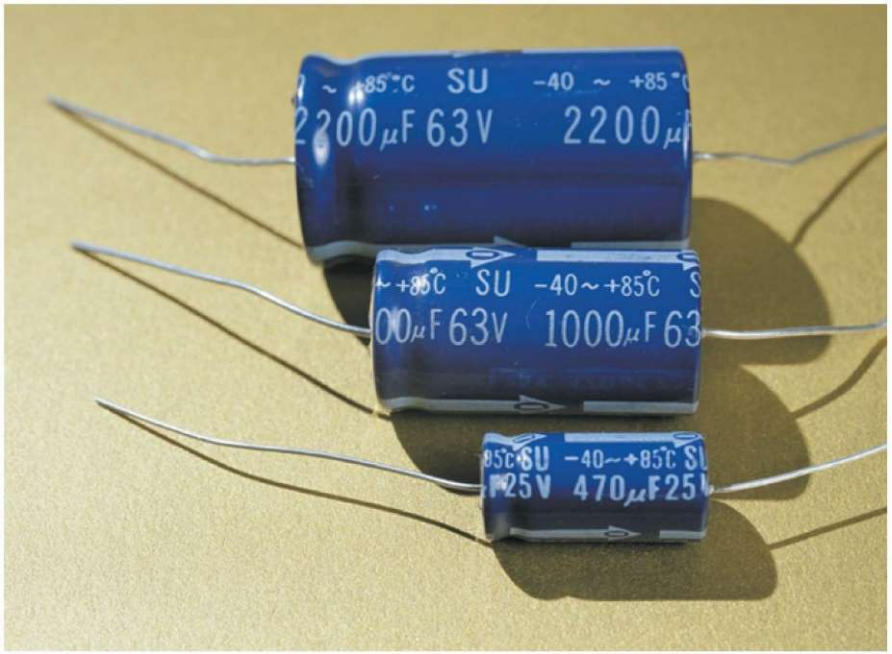
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Units of Capacitance

- The SI unit of capacitance is the farad, F.

$$1\,\text{F} = 1\,\text{C/V} = 1\,\text{C}^2/\text{N}\cdot\text{m} = 1\,\text{C}^2/\text{J}$$

- One farad is a very large capacitance.
- For the commercial capacitors shown in the photograph, C is measured in microfarads ($1\ \mu\text{F} = 10^{-6}\ \text{F}$)



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Example

The parallel plates of a 1.0-F capacitor are 1.0 mm apart. What is their area?

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Example

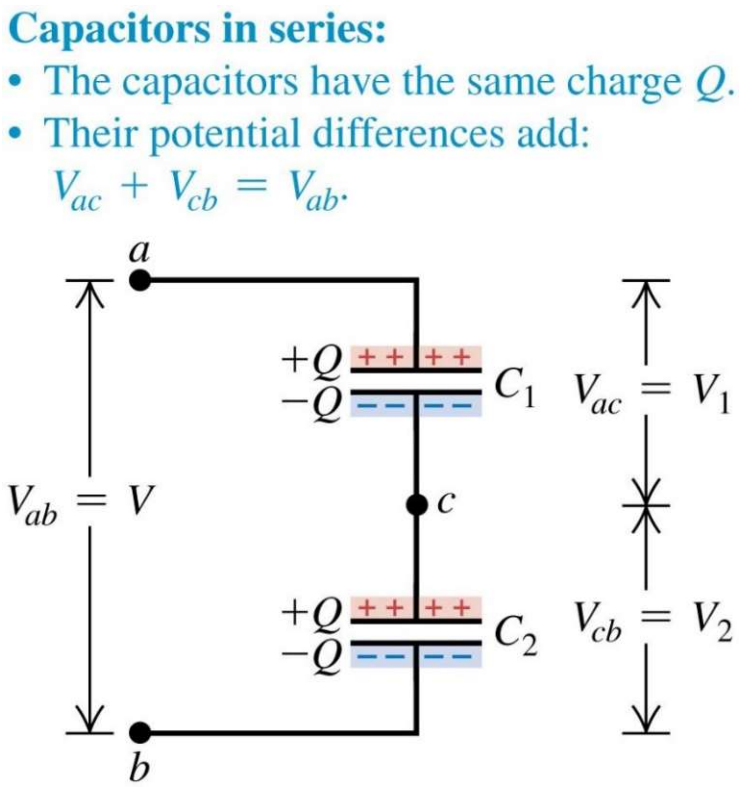
The plates of a parallel-plate capacitor in vacuum are 5.00 mm apart and 2.00 m^2 in area. A 10.0-kV potential difference is applied across the capacitor. Compute (a) the capacitance; (b) the charge on each plate; and (c) the magnitude of the electric field between the plates.

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Capacitors in Series (1 of 3)

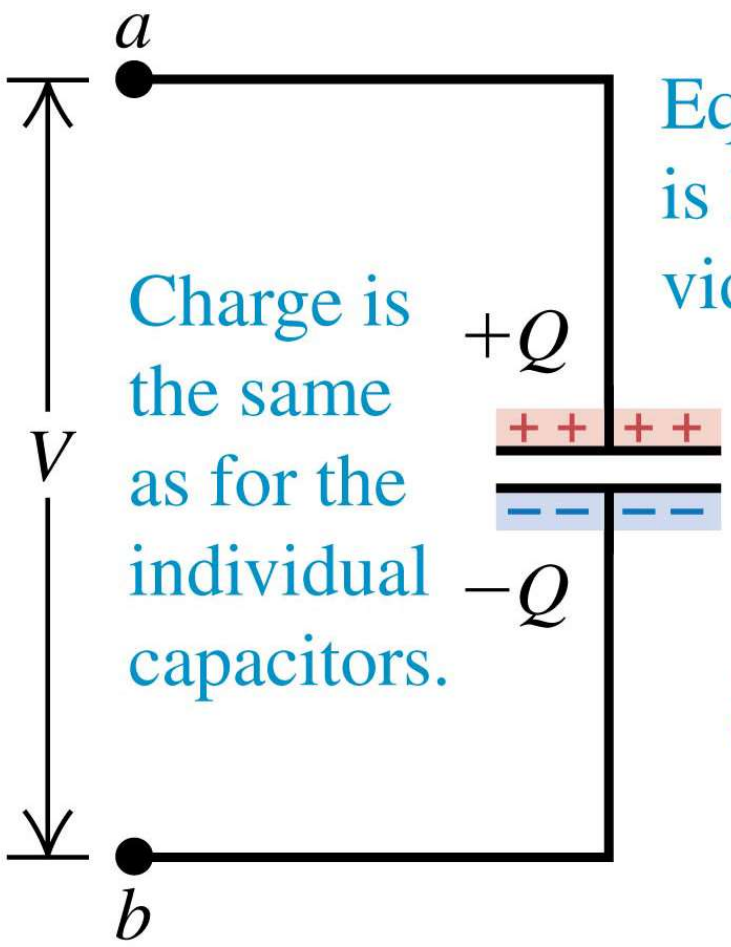
- Capacitors are in **series** if they are connected one after the other, as illustrated.
- The **equivalent single capacitor** is shown on the next slide.



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Capacitors in Series (2 of 3)



Charge is the same as for the individual capacitors.

Equivalent capacitance is less than the individual capacitances:

$$C_{eq} = \frac{Q}{V}$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

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Capacitors in Series (3 of 3)

- When several capacitors are connected in series, the magnitude of charge is the same on all plates of all the capacitors.
- The potential differences of the individual capacitors add to give the total potential difference across the series combination: $V_{\text{total}} = V_1 + V_2 + V_3 + \ldots$
- The equivalent capacitance of the series combination is given by:

Capacitors in series:

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots$$

Equivalent capacitance of series combination

Capacitances of individual capacitors

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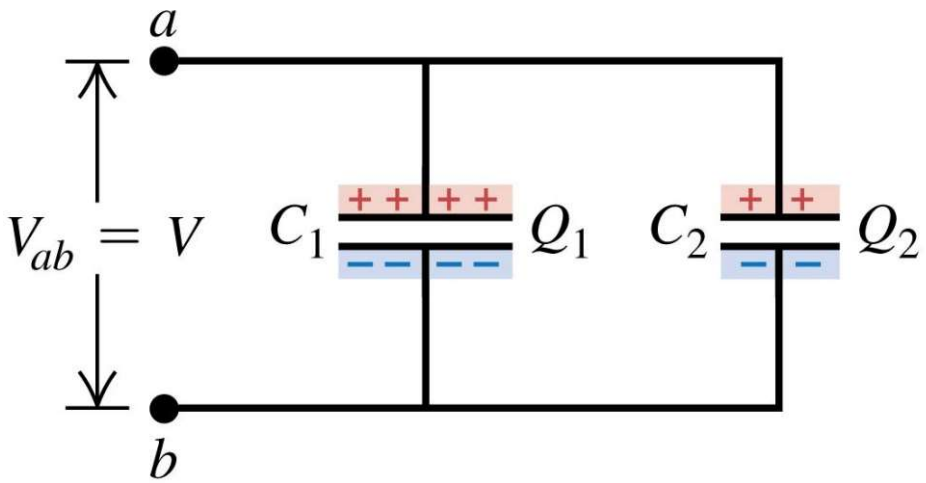
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Capacitors in Parallel (1 of 3)

- Capacitors are connected in **parallel** between *a* and *b* if the potential difference V_{ab} is the same for all the capacitors.

Capacitors in parallel:

- The capacitors have the same potential V.
- The charge on each capacitor depends on its capacitance: $Q_1 = C_1V, Q_2 = C_2V$.

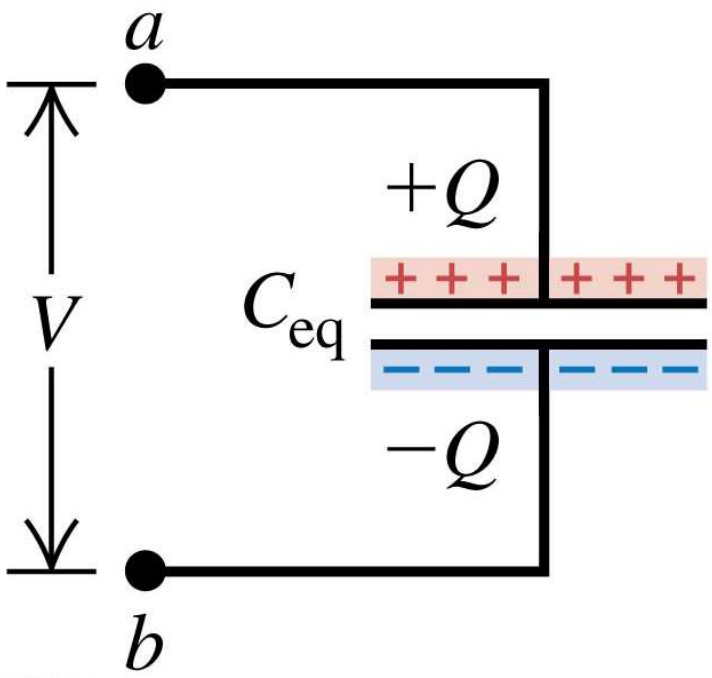


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Capacitors in Parallel (2 of 3)

- This is the equivalent capacitor of two capacitors connected in parallel.



Charge is the sum of the individual charges:

$$Q = Q_1 + Q_2$$

Equivalent capacitance:

$$C_{eq} = C_1 + C_2$$

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Capacitors in Parallel (3 of 3)

- When several capacitors are connected in parallel, the potential differences are the same for all the capacitors.
- The charges on the individual capacitors add to give the total charge on the parallel combination:

$$Q_{\text{total}} = Q_1 + Q_2 + Q_3 + \ldots$$

- The equivalent capacitance of the parallel combination is given by:

Capacitors in parallel:

$$C_{\text{eq}} = C_1 + C_2 + C_3 + \cdots$$

C_{eq}: Equivalent capacitance of parallel combination

$C_1 + C_2 + C_3 + \cdots$: Capacitances of individual capacitors

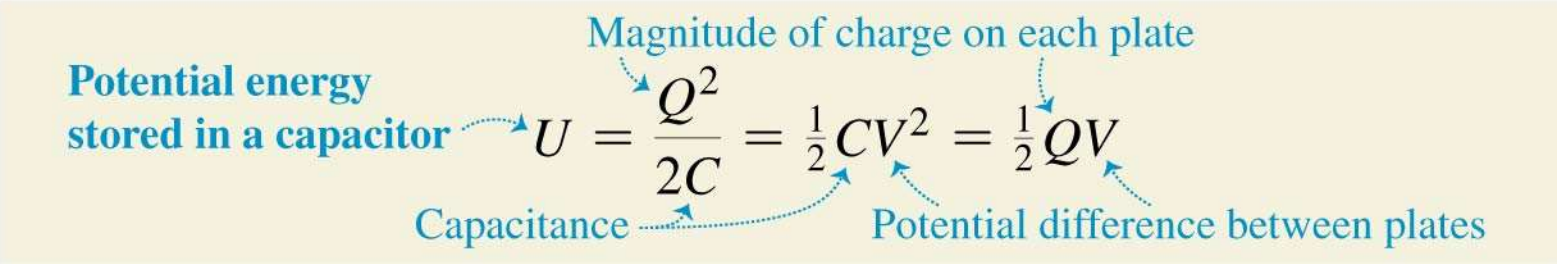
- Video Tutor Demonstration: Discharge Speed for Series and Parallel Capacitors

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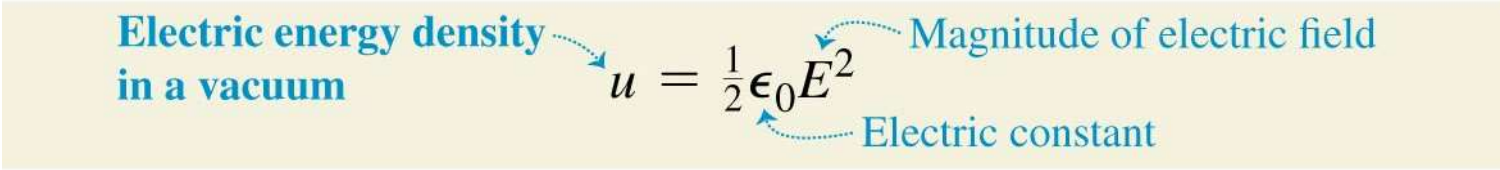
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Energy Stored in a Capacitor (1 of 2)

- The potential energy stored in a capacitor is:



- The capacitor energy is stored in the **electric field** between the plates.
- The **energy density** is:



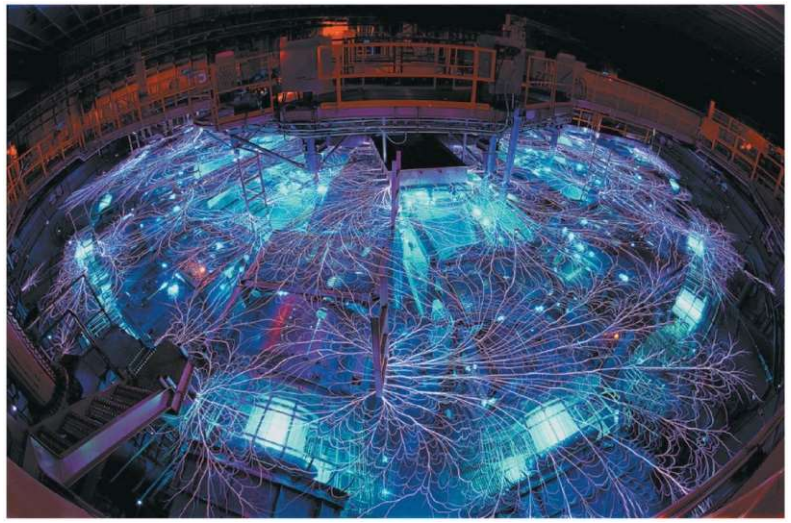
- Video Tutor Solution: Example 24.8

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Energy Stored in a Capacitor (2 of 2)

- A practical application of capacitors is their ability to store energy and release it quickly.
- An extreme example of the same principle is the Z machine at Sandia National Laboratories in New Mexico, which is used in experiments in controlled nuclear fusion.
- The Z machine uses a large number of capacitors in parallel to give a tremendous equivalent capacitance.
- The arcs shown here are produced when the capacitors discharge their energy into a target, which is heated to a temperature higher than 2×10^9 K.

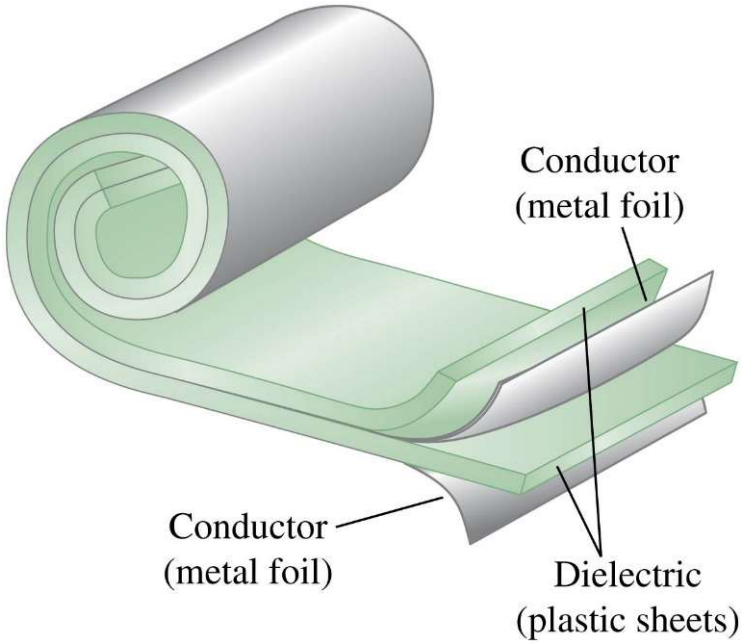


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Dielectrics (1 of 2)

- Most capacitors have a nonconducting material, or dielectric, between their conducting plates.
- A common type of capacitor uses long strips of metal foil for the plates, separated by strips of plastic sheet such as Mylar.

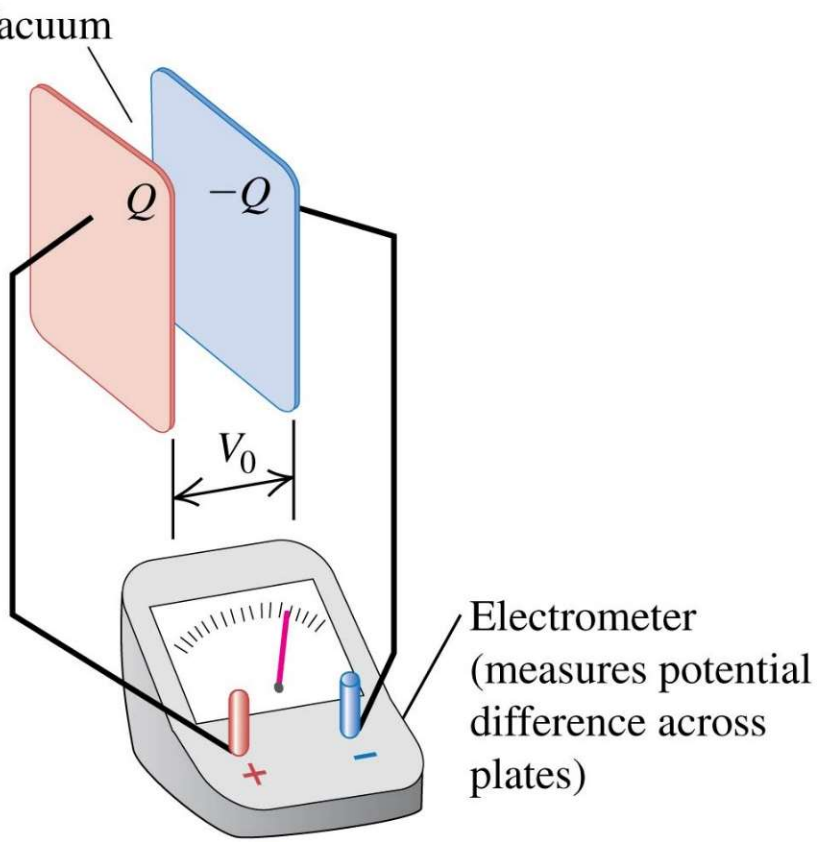


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Dielectrics Increase Capacitance (1 of 2)

- Consider an electrometer connected across a charged capacitor, with magnitude of charge Q on each plate and potential difference V_0.

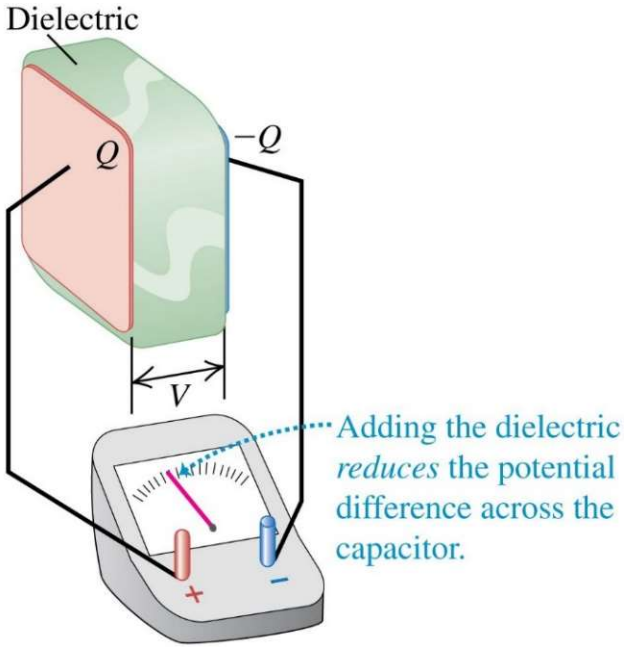


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Dielectrics Increase Capacitance (2 of 2)

- When we insert an uncharged sheet of dielectric, such as glass, paraffin, or polystyrene, between the plates, the potential difference decreases to a smaller value V.
- Since Q is unchanged, the capacitance $C = \frac{Q}{V}$ is **increased** with the dielectric.

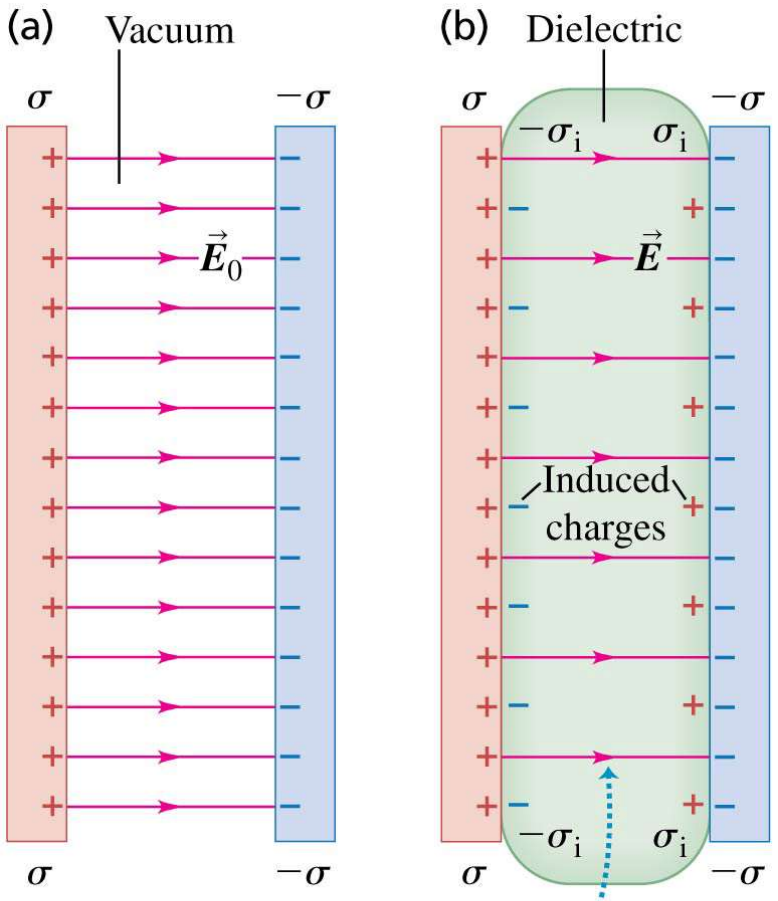


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Dielectrics (2 of 2)

- When a dielectric is inserted between the plates of a capacitor, the electric field **decreases**.
- This is due to **polarization** of the charge within the dielectric, which results in induced surface charges, as shown.



For a given charge density σ, the induced charges on the dielectric's surfaces reduce the electric field between the plates.

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The Dielectric Constant

- When an insulating material is inserted between the plates of a capacitor whose original capacitance is C_0, the new capacitance is greater by a factor K, where K is the **dielectric constant** of the material.

Capacitance of a parallel-plate capacitor, dielectric between plates

$$C = KC_0 = K\epsilon_0\frac{A}{d} = \epsilon\frac{A}{d}$$

Dielectric constant; Area of each plate; Permittivity $= K\epsilon_0$; Capacitance without dielectric; Electric constant; Distance between plates

- The energy density in the capacitor also increases:

Electric energy density in a dielectric

$$u = \tfrac{1}{2}K\epsilon_0E^2 = \tfrac{1}{2}\epsilon E^2$$

Dielectric constant; Permittivity $= K\epsilon_0$; Electric constant; Magnitude of electric field

- Video Tutor Solution: Example 24.11

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Table 24.1 Some Dielectric Constants

Material	*K*
Vacuum	1
Air (1 atm)	1.00059
Teflon	2.1
Mylar	3.1
Glass	5 – 10
Glycerin	42.5
Water	80.4

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Dielectric Breakdown

- If the electric field is strong enough, **dielectric breakdown** occurs and the dielectric becomes a conductor.
- The **dielectric strength** is the maximum electric field the material can withstand before breakdown occurs.
- For example, Pyrex glass has a dielectric constant of $K = 4.7$, and a dielectric strength of $E_m = 1 \times 10^7$ V/m.
- Dry air has a dielectric constant of $K = 1.00059$ and a dielectric strength of $E_m = 3 \times 10^6$ V/m.

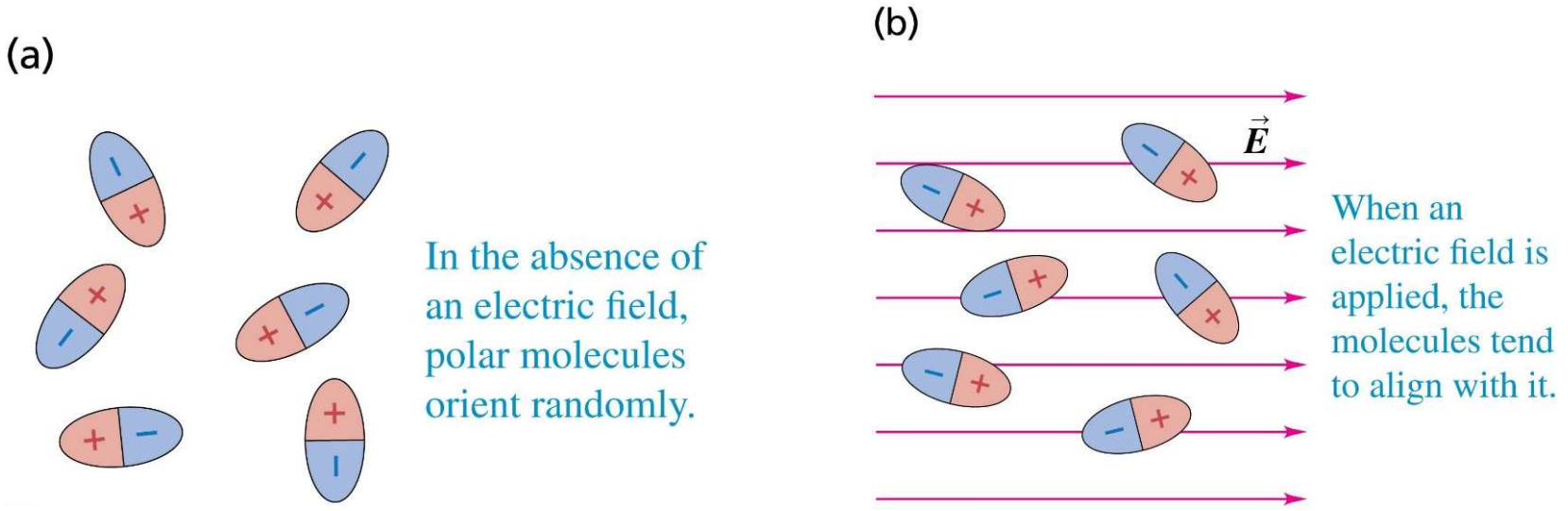


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Molecular Model of Induced Charge (1 of 3)

- Some molecules, such as H_2O and N_2O, form natural electric dipoles, and the molecule is called a **polar molecule**.
- When no electric field is present in a gas or liquid with polar molecules, the molecules are oriented randomly (a).
- In an electric field, however, they tend to orient themselves as in (b).



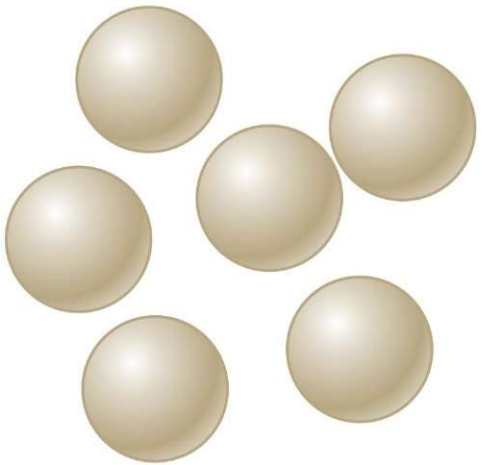
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Molecular Model of Induced Charge (2 of 3)

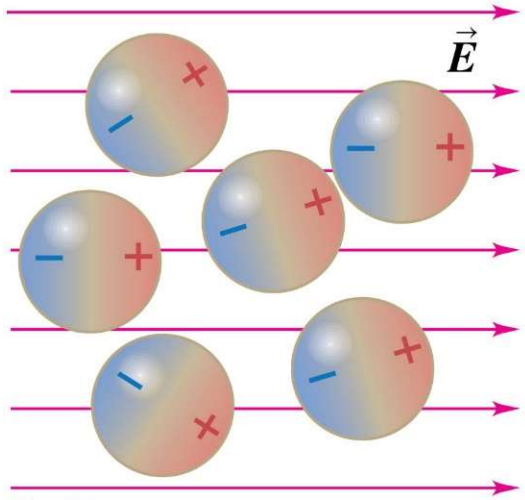
- Even a molecule that is not ordinarily polar (a) becomes a dipole when it is placed in an electric field because the field pushes the positive charges in the molecules in the direction of the field and pushes the negative charges in the opposite direction (b).
- Such dipoles are called **induced** dipoles.

(a)



In the absence of an electric field, nonpolar molecules are not electric dipoles.

(b)



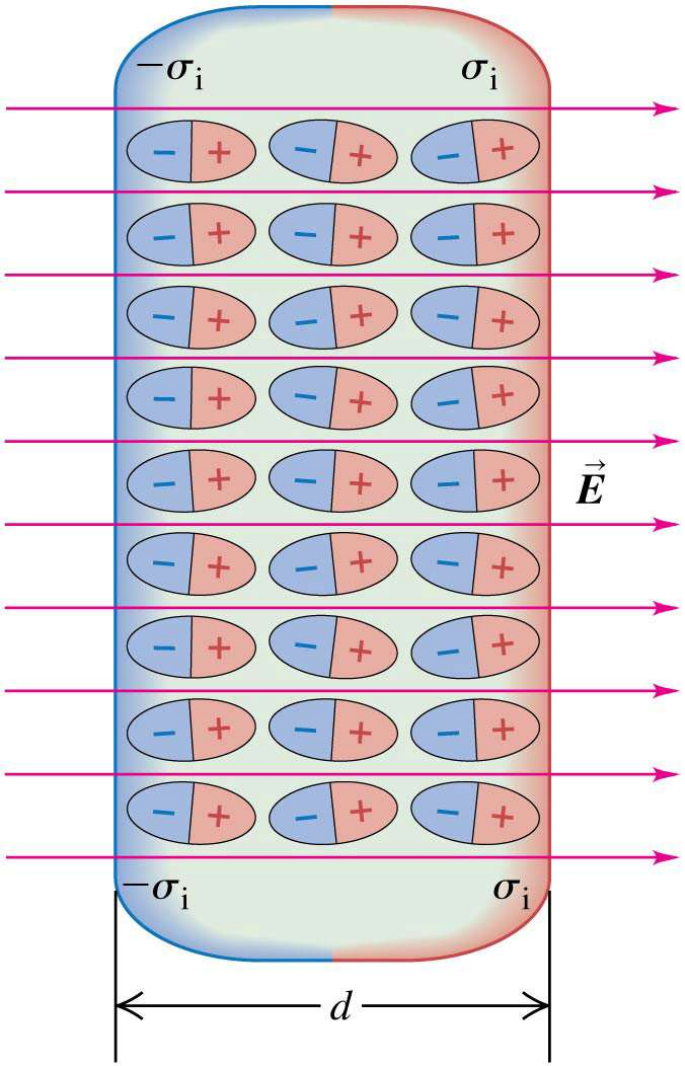
An electric field causes the molecules' positive and negative charges to separate slightly, making the molecule effectively polar.

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Molecular Model of Induced Charge (3 of 3)

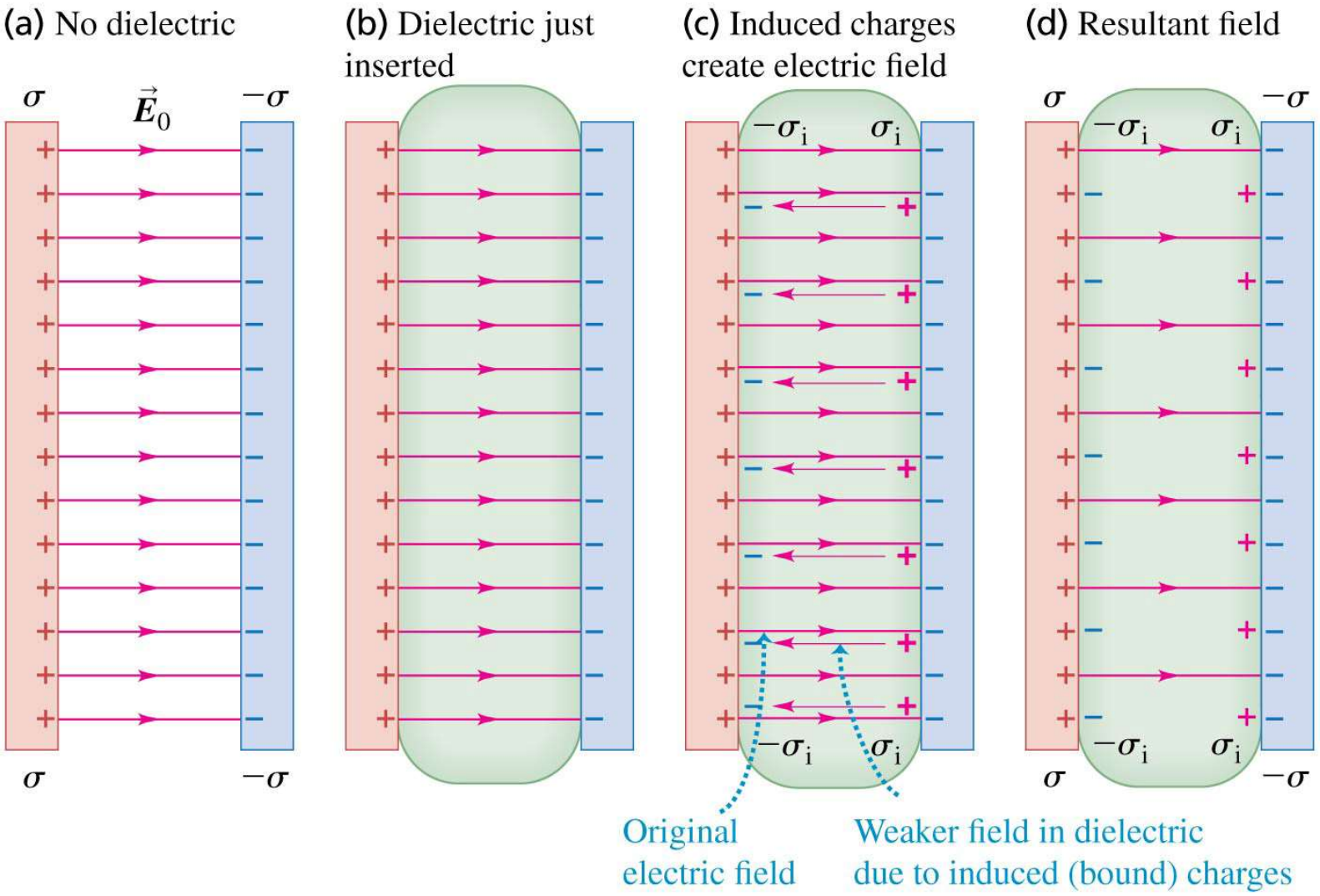
- The polarization of molecules within a dielectric leads to the formation of a layer of charge on each surface of the dielectric material.
- These layers have a surface charge density of magnitude σ_i.



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Behavior of a Dielectric in Four Steps

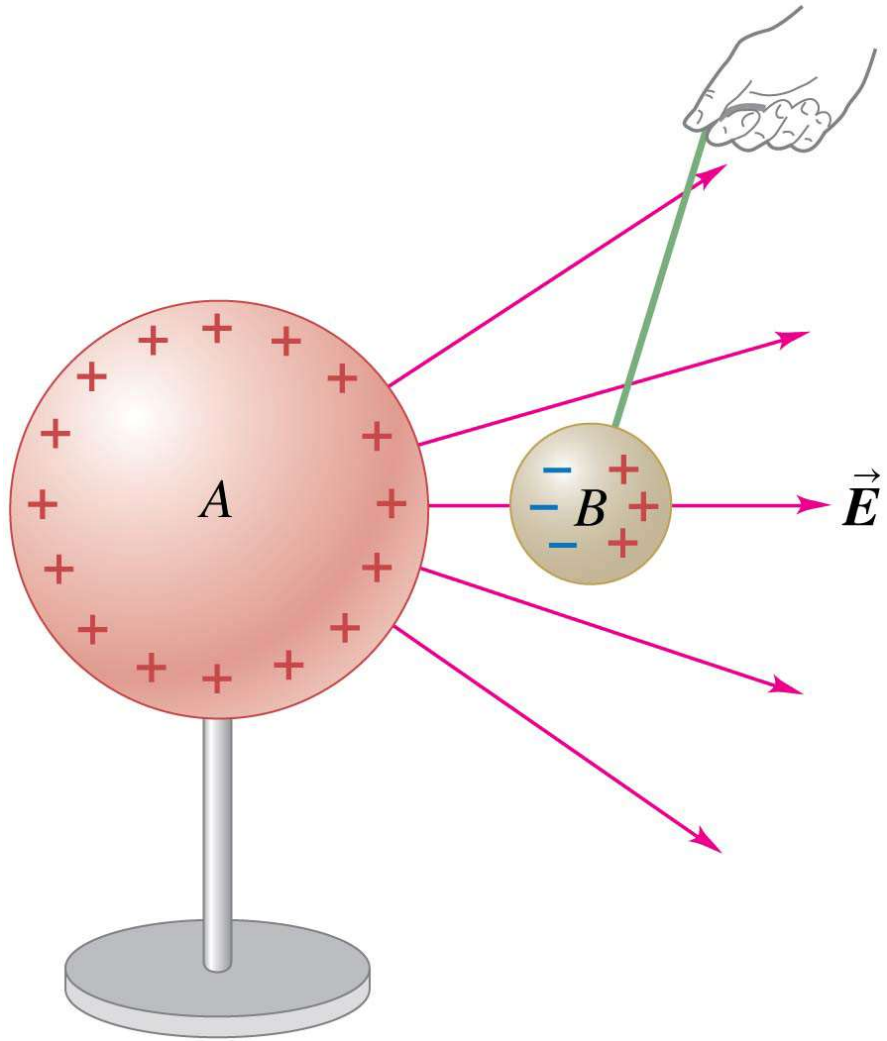


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Neutral Sphere Attracted to a Charged Sphere

- Polarization is the reason a charged object can exert a force on an uncharged nonconducting object such as a bit of paper or a pith ball.
- The force is always attractive.

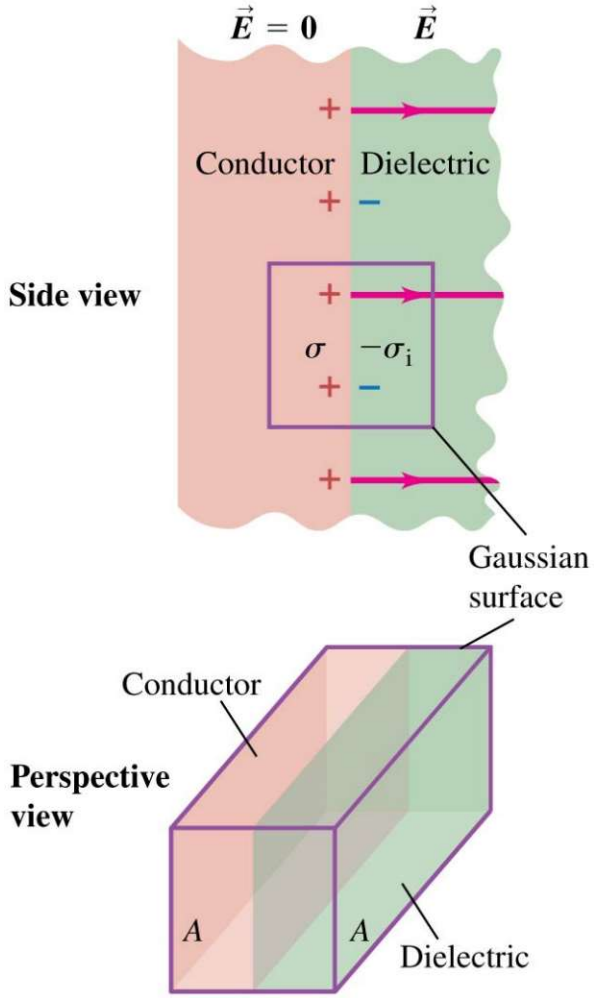


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Gauss's Law in Dielectrics

- The figure shows a close-up of the left-hand plate in a capacitor.
- The Gaussian surface is a rectangular box that lies half in the conductor and half in the dielectric.
- Gauss's law becomes:



Gauss's law in a dielectric:

$$\oint K\vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl-free}}}{\epsilon_0}$$

Dielectric constant

Total free charge enclosed by surface

Surface integral of $K\vec{E}$ over a closed surface

Electric constant

where $Q_{\text{encl-free}}$ is the total **free** charge (not bound charge) enclosed by the Gaussian surface.

- Video Tutor Solution: Example 24.12

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End of Chapter

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